

Does a regular background bias 2D causal-set lattice-gas quantum gravity? A random-background comparison

Independent project note, v1 (2026-10-09) -- not peer reviewed

This is a short note. For full validation results, every pitfall found and fixed along the way, and commit-level citations for every number below, see `full_report.md` in this same folder.

The open question

Cunningham & Surya's dimensionally-restricted "lattice gas" [arXiv:1908.11647] makes 2D causal-set Monte Carlo cheap enough to run by placing causal-set elements on a fixed *regular* lattice embedded in a cylinder spacetime, rather than sampling generic partial orders directly. It reproduces the generic model's phase transition between a manifold-like phase and a layered, highly-ordered phase at much lower computational cost. In their Conclusions (Sec. 5), C&S write: "we have chosen a specific, regular lattice, and we don't yet know whether our results will change with a different lattice geometry. A more natural choice for the background lattice is a random m element causal set obtained via a sprinkling into (M, g) . It would be important to explore simulations on such lattices and compare with our current results" -- flagged as future work, not pursued in their paper.

The answer

Yes, the location of the transition depends on the background being regular -- the effect is real, significant, and shrinks with system size, but does not vanish at the two sizes tested; the qualitative physics (phase structure, hot-phase statistics) does not depend on the background.

We built a random-background variant of the lattice gas: identical region, filling representation, Monte Carlo moves, and action, with the only change being that the m lattice sites are replaced by a fixed (quenched) uniform-random sprinkling of continuous positions. An explicit equivalence test confirms this reduces exactly to the regular lattice's own causal structure when the random positions are placed back on the integer grid, so "random background" and "regular lattice" differ only in that one respect.

What we measured

Using a Wang-Landau / multicanonical sampler (built to replace an earlier parallel-tempering scan whose results turned out to be mostly right-censored -- see `full_report.md` Section 6), we located the critical coupling β_c at two system sizes, $n=30$ and $n=50$, for both backgrounds. "5 seeds" means different things for the two backgrounds: for the random background, 5 independent background realizations (5 different quenched sprinklings, each with its own Monte Carlo chain); for the regular lattice, which is the same deterministic lattice every time, 5 independent Monte Carlo chain seeds on that one lattice:

n	Random background β_c	Regular lattice β_c	Gap	z	Relative shift
30	9.723 +/- 0.026 (SEM)	11.095 +/- 0.005 (SEM)	1.372	51.8	12.4%
50	5.959 +/- 0.033 (SEM)	6.419 +/- 0.005 (SEM)	0.460	13.67	7.2%

At both sizes the regular lattice's located β_c is significantly higher than the random background's -- not explainable by run-to-run noise on either side ($z=51.8$ and $z=13.67$ are both far beyond any conventional significance threshold). The regular-lattice value at $n=30$ is also cross-checked against this project's independent earlier measurement of the same model (within about 2 standard deviations), and the random background's result is robust to the exact pipeline used to produce it: two different Wang-Landau sampling methods agree to within 0.008, checked by directly re-running one seed from each background (one random-background realization, one regular-lattice chain seed) under both methods.

The *relative* gap shrinks with size, consistent with (not proven by) $1/n$ scaling -- the same rate β_c itself falls off here. The relative shift times n is close to constant across the two sizes we tested ($12.4\% \times 30 = 371$ vs. $7.2\% \times 50 = 358$, a two-point power-law exponent of about 1.07). Because the *absolute* gap is the relative gap times β_c , and both factors fall off at roughly $1/n$, the absolute gap falls off faster, close to $1/n^2$: it drops 2.98x from $n=30$ to $n=50$ ($1.372 \rightarrow 0.460$), close to the $(50/30)^2 = 2.78$ x a pure $1/n^2$ law predicts, not the 1.67x a $1/n$ law on the absolute gap alone would give. Taken at face value, the $1/n$ relative-gap reading predicts a relative shift of about 5.6% at $n=64$ -- the largest size reachable with this project's current code without a bitset-engine rewrite (every model here caps out at $N, n \leq 64$) -- a concrete number anyone can check against a direct run. Extrapolating further (well beyond where this project has any data, so treat this as a back-of-envelope number, not a result): at $n=200$, the system size C&S themselves use for their 2D lattice gas, the same $1/n$ reading predicts a relative gap of only about **1.8%** -- small enough that it's plausible their published 2D results are only mildly affected by the regular background, and directly testable by running their own code on a sprinkled background at their own system size.

We checked and ruled out the simplest mechanism. A natural guess is that the random background simply reaches a deeper, more-ordered cold-phase action floor, which would mechanically pull β_c down. Directly comparing the lowest action reached and the hot-phase mean action between the two backgrounds shows no statistically significant difference -- this hypothesis does not hold. The real mechanism (presumably something about the shape of the density of states across the explored range, not its extremes) is still open.

What does *not* change: the hot phase ($\beta=0$ fillings) is close to background-independent at both sizes -- every measured observable (Myrheim-Meyer dimension, ordering fraction, height, occupied row count) lands within about one standard deviation of the regular lattice's own value. The cold phase differs more at $n=30$ but that gap has nearly closed by $n=50$ (ordering fraction 0.661 vs. 0.660, MM dimension 3.60 vs. 3.60). Neither background, at either size, shows a resolved double-peaked action histogram at its own critical coupling -- so this project does not confirm a first-order transition on either background, and the random-background result above is about *where* the transition sits, not about resolving its order.

Caveats

- **Only two system sizes.** The $1/n$ -consistent shrinkage of the *relative* gap above is a two-point observation, not a fitted law; a third size ($n=64$ is the concrete next test) would meaningfully sharpen this.
- **Mixed sampler methods at $n=50$.** Two Wang-Landau bugs were found and fixed partway through the $n=50$ runs (an unphysical histogram cliff from a flat-fill widening step, and a genuinely kinetically-isolated bin that no amount of reweighting could repair -- both described in full in `full_report.md` Section 6). The $n=50$ dataset mixes 5 seeds run under the original method with 5 run under the fixed method; a direct 2-seed consistency check found no material difference (0.002 and 0.008), but a fully single-method $n=50$ dataset was not produced.
- **Mechanism still open.** We ruled out the simplest explanation for *why* the backgrounds differ, not the real cause.
- **The lattice causal relation is our own derivation, not a quote.** C&S's paper states 45-degree lightcones and cylinder wraparound but does not give an explicit integer-coordinate formula, so we derived one ourselves (shortest arc on the circumference) and flagged it explicitly as our own construction rather than a quote. Supporting evidence that it matches their intended model: our regular-lattice hot phase reproduces their own quoted ordering fraction almost exactly (0.882 here vs. their 0.88), which would be a coincidence if the causal relation were substantially wrong.
- **2D only.** C&S's own paper, and the underlying causal-set action, both extend to higher dimensions; this project has not run a 3D or 4D version of either model.

Where to go next

`full_report.md` in this same folder has the complete validation story (the order model checked against the published finite-size- scaling formula, the lattice gas checked against C&S's own quoted numbers), every methodological pitfall found and fixed along the way, and exact commit-level citations for every number in this note. The code, tests, and scripts to reproduce any of it are in the same repository; see `README.md` at the repository root.